

CASA GRANDE, AZ, USA

WMO: 722748

Lat: 32.950N

Lon: 111.767W

Elev: 446

StdP: 96.09

Time Zone: -7.00 (W07)

Period: 94-19

WBAN: 03914

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	-0.2	1.9	-16.1	1.0	7.2	-13.0	1.3	10.2	10.2	14.6	8.8	15.3	1.2	50	0.447

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.6	42.5	20.9	41.4	20.7	40.5	20.5	23.6	33.7	23.1	34.1	22.7	34.1	4.0	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.4	16.9	26.7	20.5	16.0	27.3	19.0	14.6	28.3	72.7	33.9	70.9	33.9	69.1	33.9	26.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.2	8.0	6.9	DB	-3.2	44.6	1.5	1.2	-4.4	45.5	-5.3	46.2	-6.1	46.9	-7.3	47.7	(4)
(5)			WB	-5.8	24.4	1.9	0.7	-7.1	24.9	-8.3	25.3	-9.3	25.6	-10.7	26.1	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	21.6	11.4	12.8	16.6	20.1	24.9	30.2	32.9	32.2	29.1	22.2	15.6	10.9
	DBStd	8.52	3.22	3.41	3.57	3.83	3.59	2.98	2.29	2.25	2.70	3.90	3.63	3.11
	HDD10.0	67	22	11	1	0	0	0	0	0	0	0	4	28
	HDD18.3	814	216	156	77	26	2	0	0	0	0	12	94	232
	CDD10.0	4306	65	89	204	302	462	607	710	688	573	377	173	54
	CDD18.3	2013	1	1	22	78	205	357	452	430	323	131	13	0
	CDH23.3	31373	12	63	464	1268	3175	5970	7364	6733	4482	1594	242	6
	CDH26.7	19589	0	5	136	568	1854	4066	5027	4427	2692	759	56	0
Wind	WSAvg	2.4	2.1	2.2	2.4	2.9	2.9	2.7	2.9	2.6	2.4	2.3	2.1	2.0
Precipitation	PrecAvg	222	19	17	22	7	3	3	22	47	20	20	19	30
	PrecMax	386	76	57	78	37	16	24	84	158	50	94	62	120
	PrecMin	124	0	0	0	0	0	0	2	2	0	0	0	0
	PrecStd	72	18	18	20	10	5	6	20	41	18	22	18	35
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	25.0	27.4	32.4	36.2	41.0	43.0	44.1	43.0	41.1	37.4	31.1	24.0
		MCWB	11.4	12.4	14.8	16.3	18.9	20.0	21.5	21.3	20.7	18.3	14.8	11.3
	2%	DB	22.6	25.8	29.9	33.8	38.0	42.2	42.6	42.1	39.1	35.2	28.0	22.2
		MCWB	10.7	11.9	13.9	15.1	17.4	19.9	21.3	21.5	20.2	17.7	13.3	11.1
	5%	DB	21.1	22.9	27.8	32.3	37.1	41.1	41.9	40.9	37.9	32.9	26.3	19.9
		MCWB	10.1	10.8	12.9	14.6	17.0	19.4	21.2	21.6	19.9	16.5	12.8	10.2
	10%	DB	18.8	21.1	26.2	30.2	35.1	39.8	40.2	39.0	37.1	31.2	23.9	17.8
		MCWB	9.4	10.3	12.2	13.8	15.8	18.9	21.0	21.4	19.8	16.1	11.7	9.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.8	14.1	15.6	16.7	19.1	22.2	24.4	24.3	23.5	21.6	15.8	14.1
		MCDB	18.6	21.3	29.4	34.1	38.5	36.9	32.7	33.6	32.3	27.2	24.6	19.2
	2%	WB	12.4	13.1	14.4	15.7	18.0	21.3	23.7	23.7	22.8	19.7	14.6	12.6
		MCDB	18.6	20.9	27.2	32.6	36.6	36.9	33.5	34.0	32.1	30.6	24.3	18.2
	5%	WB	11.3	12.1	13.6	14.8	17.2	20.4	23.1	23.2	22.2	18.4	13.6	11.3
		MCDB	18.1	20.5	25.9	30.9	35.3	37.1	34.4	34.3	31.9	29.3	23.7	17.5
	10%	WB	10.2	11.1	12.7	14.0	16.4	19.6	22.7	22.8	21.6	17.2	12.7	10.2
		MCDB	17.4	19.6	24.6	29.3	34.1	37.6	34.6	34.3	31.8	28.7	22.9	16.6
Mean Daily Temperature Range	5% DB	MDBR	14.0	14.2	15.9	16.7	17.2	17.6	13.6	13.1	14.0	15.3	15.0	13.6
		MCDBR	16.7	17.7	18.3	19.2	19.0	18.3	15.8	15.0	15.4	17.3	17.5	16.3
		MCWBR	9.0	8.9	8.5	8.8	8.6	7.7	5.5	4.6	5.4	7.5	8.9	8.8
	5% WB	MCDBR	13.2	14.9	16.9	18.0	18.0	15.3	12.1	12.3	11.8	13.6	15.1	13.2
		MCWBR	7.4	7.7	7.7	8.3	8.0	5.8	3.4	3.4	3.9	6.0	7.9	7.4
Clear-Sky Solar Irradiance	taub	0.275	0.280	0.305	0.314	0.322	0.336	0.396	0.383	0.342	0.314	0.285	0.275	
	taud	2.563	2.556	2.461	2.427	2.434	2.400	2.360	2.386	2.469	2.495	2.533	2.562	
	Ebn at Noon	957	985	976	974	961	943	885	894	923	930	935	938	
	Edh at Noon	80	89	106	115	115	119	123	118	104	94	81	76	
All-Sky Solar Radiation	RadAvg	3.47	4.43	5.92	7.32	8.17	8.40	7.28	6.74	6.08	4.99	3.84	3.15	
	RadStd	0.25	0.33	0.27	0.20	0.27	0.35	0.34	0.21	0.24	0.28	0.17	0.15	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (14 neighbors)	+0.31	N/A	N/A	N/A	N/A	N/A	N/A	-84	+109	N/A

Nomenclature: See separate page